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CMR INSTITUTE OF TECHNOLOGY, BENGALURU FINANCIAL ACCOUNTING & REPORTING

MBA102 (Batch 2024-26)

MODULE NOTES VI

Subject Teacher : Dr G Rajasekar

Depreciation: Meaning, characteristics and causes of depreciation, Types of Depreciation. Tax implication of depreciation. (Problems only on straight line and WDV method).

Emerging Issues in Accounting: Human Resource Accounting, Forensic Accounting, Green Accounting, Sustainability Reporting, Automated Accounting Processes, Cloud-based Accounting, Data Analytics & Forecasting Tools, Blockchain, AI in Accounting, Big Data in Accounting (Theory only).

Meaning, characteristics and causes of depreciation

Meaning of Depreciation

Depreciation can be defined as a continuing, permanent and gradual decrease in the book value of fixed assets. This type of shrinkage is based on the cost of assets utilised in a firm and not on its market value.

Features of Depreciation

Following are the 3 principal features of depreciation:

- ⑩ Depreciation is a decrease in the book value of fixed assets.
- ⑩ Depreciation involves loss of value of assets due to the passage of time and obsolescence.
- ⑩ Depreciation is an ongoing process until the end of the life of assets.



DEPRECIATION

Causes of Depreciation

1. **Wear and Tear due to Use or Passage of Time:** Wear and tear is nothing but deterioration and the following decrease in the value of an asset, resulting from its use in business operations for earning revenue.
2. **Expiration of Legal Rights:** Some categories of assets lose their value after the agreement directing their use in business comes to an end after the expiry of the predetermined period.
3. **Obsolescence:** Obsolescence is another factor driving to the depreciation of fixed assets. In common language, obsolescence means being “out-of-date”. Obsolescence refers to an actual asset becoming outdated on account of the availability of a better type of asset.
4. **Abnormal Factors:** Drop in the use of the asset may be caused by abnormal factors. Namely, accidents due to the earthquake, fire, floods, etc., Accidental loss is permanent but not continuing.

Depreciation Formula:

Depreciation Methods		
Straight-Line Depreciation	=	$\frac{\text{Cost of Asset - Residual Value}}{\text{Useful Life of an Asset}}$
Double-Declining Balance Depreciation	=	$\frac{\text{Cost of Asset} \times \text{Rate of Depreciation}}{100}$
Units-of-Production Depreciation	=	$\frac{\text{Cost of Asset - Salvage Value}}{\text{Useful Life in the Form of Units Produced}}$
Sum-of-the-Years' Digits Depreciation	=	$\frac{\text{Remaining Useful Life of Asset}}{\text{Sum of the Year's Digit}} \times \text{Depreciable Cost}$

Depreciation Formula

$$\text{Straight Line Depreciation Method} = \frac{\text{Cost of an Asset} - \text{Residual Value}}{\text{Useful life of an Asset}}$$

$$\text{Diminishing Balance Method} = \frac{\text{Cost of an Asset} \times \text{Rate of Depreciation}}{100}$$

$$\text{Unit of Product Method} = \frac{\text{Cost of an Asset} - \text{Salvage Value}}{\text{Useful life in the form of Units Produced}}$$

Advantages and Disadvantages of Straight Line Method:

Advantages	Disadvantages
1. It is a very simple method of calculating depreciation.	1. Under this method book value of the asset will be charged more for maintenance and repair in the final years as compared to initial years.
2. Under this method, Asset can be depreciated up to the net scrap value or zero value.	2. It is difficult to ascertain a suitable rate of depreciation.
3. Under this method, the same amount is charged as depreciation in Profit & Loss Account.	3. It is not suitable for assets having long life and high value.

Methods of Calculating Depreciation

Straight Line Method (SLM)

Under the depreciation Straight Line Method, a fixed depreciation amount is charged annually, during the lifetime of an asset. The amount of annual depreciation is computed on Original Cost and it remains fixed from year to year. This method is also known as the 'Original Cost method' or 'Fixed Instalment method'.

Written Down Value Method (WDV)

Under the Written Down Value method, depreciation is charged on the book value (cost -depreciation) of the asset every year. Under the WDV method, book value keeps on reducing so, annual depreciation also keeps on decreasing. This method is also known as 'Diminishing Balance Method' or 'Reducing Instalment Method'.

Straight Line Method vs Written Down Value Method

Straight Line Method	Written Down Value Method
Charging Depreciation	Basis of
Depreciation is calculated on the original cost of fixed assets.	Depreciation is calculated on the book value of fixed assets.
Annual Depreciation	Amount of
The amount of annual depreciation is fixed for all years of useful life.	The amount of depreciation declines year after year.
by Income Tax	Recognition
The Straight Line Method is not	The Written Down Value method is

recognised by the Income Tax Department.	recognised by the Income Tax Department.
Depreciation and Repairs	Cost of
The combined cost on account of depreciation and repairs is lower in the initial years and higher in the later years.	The combined cost on account of depreciation and repairs remains, more or less, equal throughout the life of the asset.

Theory Questions:

Q.1- What are the other concepts similar to depreciation?	
Answer:	
Depletion	⑩ The term 'Depletion' is used with regard to the 'Natural Resources' like oil wells, Mines etc. ⑩ When natural resources are extracted and their stock value is reduced. This reduction is termed as depletion.
Amortisation	⑩ The term 'Amortisation' is used with regard to 'Intangible Assets'. ⑩ Amortisation refers to writing off the cost of intangible assets like patents, copyright, trademarks, franchises, etc.
Obsolescence	⑩ It refers to a decline in the value of assets due to innovation or improved techniques, changes in the taste or fashion of the existing asset.

Q.2- State the objectives of depreciation.

Answer:

Objectives of Depreciation	⑩	To ascertain the correct cost of production
	⑩	To retain funds for a replacement.
	⑩	To ascertain the correct Profit or Loss.
	⑩	To show a true and fair view of the financial position.

Q.3- List the factors required for calculating the amount of depreciation.

Also give the components of original cost.

Answer:

(a) Factors/elements required to calculate depreciation	⑩	The original cost of the assets.
	⑩	Estimated residual value or scrap value.
	⑩	The estimated useful life of the assets.

(b) Original cost	Original cost =	
	Purchase price + Freight + Installation cost +	
	Any other expenses incurred before the asset is put to use for the first time.	
	⑩	In the case of second-hand assets, even expenses incurred on repair & overhauling, before putting it to use, are also included in the original cost.

Multiple Choice Questions

Q.1- Amortisation refers to writing off the cost of:

- a. Fixed Assets
- b. Tangible Assets
- c. Intangible Assets
- d. All of the above

Q.2- _____refers to decline in the value of assets due to Innovation or improved techniques, changes in the taste or fashion of the existing asset.

- a. Intangible Assets
- b. Obsolescence
- c. Amortisation
- d. Depreciation

Q.3- Depreciation is a _____expense.

- a. Non-cash
- b. Cash
- c. Credit
- d. All of the above

Answer

Key

1-c, 2-b, 3-a

Emerging Issues in Accounting: Human Resource Accounting, Forensic Accounting, Green Accounting, Sustainability Reporting. (Theory only)

What is Human Resource Accounting?

Human resource accounting (HRA) is a type of accounting that seeks to determine the cost and value of the people, also known as human capital, working in an organization.

HRA is an economic indicator of what an organization spends on its human capital. It indicates money spent on recruiting, training, salary and benefits of existing employees in previous months and years.

Human Resource Accounting Definitions

American Accounting Society Committee (1973)

“Human Resource Accounting is the process of identifying and measuring data about HR and communicating this information to interested parties.”

Davidson and Roman L Weil

“HRA describes a variety of proposals that seek to report and emphasize the importance of human resources – knowledgeable, trained and loyal employees in a company earning process and total assets.”

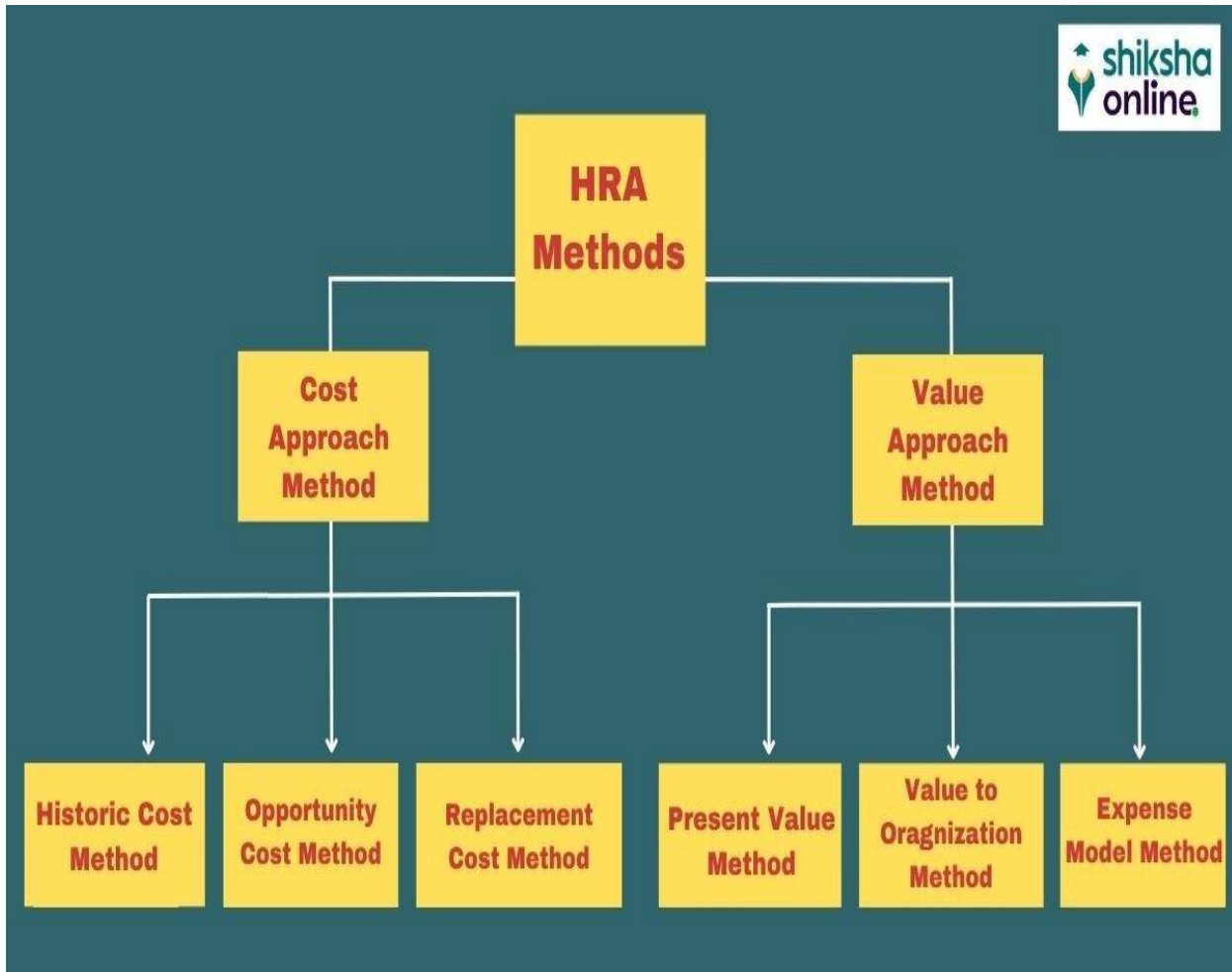
Eric Flamholtz, University of California (1974)

“Human resource accounting is the measurement of the cost and value of the people for the organisation.”

Stephen Knauf (1983)

“HRA is the measurement and quantification of human organizational inputs such as recruiting, training, experience and commitment.”

Human Resource Accounting Methods



1. Cost Approach Method

The cost approaches involve the computation of the cost of human resources to the organization. Different cost approach methods include –

- **Historical Cost Method** – The historical cost method involves calculating the cost incurred by the company in recruiting, training, and developing its employees. This approach is easy to apply but does not reflect human assets' true value.
- **Replacement Cost Method** – This method involves estimating the cost of replacing an employee with a similar level of skill and experience.
- **Opportunity Cost Method** – The opportunity cost method estimates the potential earnings an employee could have earned if they pursued a different career.



HR Generalist Roles and Responsibilities

HR Generalists are the backbone of the HR department, managing a wide range of responsibilities. From recruitment and employee relations to performance management and policy implementation, they play a crucial...read more

HR Executive Job Description, Roles and Responsibilities

Discover the job roles of an HR Executive. Learn about their crucial responsibilities in recruitment, training, compensation, and fostering a positive work culture.

2. Value Approach Method

The value of human resources is measured based on the contribution they are likely to make to the organizations during their employment. Different value approach methods include –

- Present Value Method – This method estimates the present value of the future cash flows an employee must generate for the company.
- Value to the Organization Method – In this method, the most valuable employee of the organization is determined and measured whether the organization is earning premium profits from the services of that employee and helps in finding the value of that employee.
- Expense Model Method – This method divides the employees into two categories: Decision-making and decision-execution. It then determines the actual cost incurred in both categories and whether it benefits the organization.

Forensic accounting

is the investigation of fraud or financial manipulation by performing extremely detailed research and analysis of financial information. Forensic accountants are often hired to prepare for litigation related to insurance claims, insolvency, divorces, embezzlement, fraud, skimming, and any type of financial theft.

Forensic Accounting



Preparing for Trial

Preparing for Litigation

“Forensic evidence” means evidence that is able to be presented in a court of law. Hence, forensic accounting is a term to describe an analysis of financial information that can be used to support a case in a court of law.

The process of digging through all of a company’s or individual’s financial information can take months or even years and requires a team of specialized accountants that act like detectives trying to solve a mystery.

Typically, an accounting firm will be engaged by a client either looking to defend themselves, or one looking to prosecute someone. Most medium- to large-sized firms have a forensic accounting department, which may consist of various forensic auditors.

Types of Forensic Accounting

There are various types of forensic auditing that can take place, and they are typically grouped by the types of legal proceedings that they fall under. Below are some of the most common examples:

- ⑩ Financial theft (customers, employees, or outsiders)
- ⑩ Securities fraud
- Bankruptcy
- ⑩ Defaulting on debt
- ⑩ Economic damages (various types of lawsuits to recover damages)
- ⑩ M&A related lawsuits
- ⑩ Tax evasion or fraud
- ⑩ Corporate valuation disputes
- ⑩ Professional negligence claims
- ⑩ Money laundering
- ⑩ Privacy information
- ⑩ Divorce proceedings



Forensic Accounting Careers

There is a broad range of career options that exist for accountants who want to get into forensic accounting. Depending on the client being represented and the nature of the trial, the work and responsibility can be distinctly different. For example, working on a personal divorce versus the Enron scandal would be vastly different.

The types of activities performed by these specialized accountants include investigating fraud, quantifying damages, valuing a company, or assessing tax bills.

Career paths into the position can vary, but they typically require several years of traditional audit experience at a public accounting firm. There are various professional accreditations that accountants can obtain to further a career in forensic accounting.

1. Chartered Financial Analyst
2. Certified Public Accountant
3. Chartered Alternative Investment Analyst
4. Certified Financial Planner
5. Financial Risk Manager
6. Financial Modeling & Valuation Analyst

Green Accounting

Green Accounting, also known as environmental accounting or sustainable accounting, is a system of accounting that takes into account the economic, environmental, and social costs and benefits of business activities. It involves measuring and reporting the impacts of economic activities on natural resources and the environment, in addition to traditional financial measures.

Green Accounting Objectives

The objectives of Green Accounting are to:

Integrate environmental costs and benefits into national accounts and decision-making processes.

Provide a comprehensive view of the true costs and benefits of economic activities by incorporating environmental and social considerations.

Promote sustainable development and support the transition towards a green economy.

Encourage transparency and accountability in the use of natural resources and the management of environmental impacts.

Foster stakeholder engagement and participation in environmental decision-making.

Green Accounting Components

Green Accounting comprises various components, including:

Component	Description
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<u>Environmental Management Systems (EMS)</u>	A framework for managing environmental impacts and complying with environmental regulations. An EMS involves establishing policies and procedures for environmental management, conducting regular environmental audits, and implementing continuous improvement measures.
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<u>Environmental Performance Indicators (EPI)</u>	Metrics are used to track and report on environmental performance, such as greenhouse gas emissions, energy consumption, and water use.
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EPIs enable businesses to monitor progress toward environmental goals and targets.

<u>Life Cycle Assessment (LCA)</u>	A method for evaluating the environmental impacts of a product or service throughout its entire life cycle, from raw material extraction to disposal. LCA can help businesses identify opportunities to reduce environmental impacts at all stages of the product life cycle.
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<u>Full Cost Accounting (FCA)</u>	An accounting approach that includes both the direct costs (such as materials, labor, and overhead) and indirect costs (such as environmental and social costs) of business activities. FCA can help businesses make more informed decisions by accounting for the full costs of their activities.
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<u>Environmental Reporting and Disclosure</u>	Reporting on environmental impacts and performance to stakeholders, such as investors, regulators, and customers. Environmental reporting can take the form of sustainability reports, environmental impact assessments, and other disclosures.
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<u>Environmental Auditing</u>	A systematic review of an organization's environmental performance to identify areas for improvement and compliance with environmental regulations. Environmental audits can be conducted internally or externally and can help businesses identify
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opportunities to reduce environmental impacts and comply with regulations.

Read about: Gini Coefficient

Green Accounting Types

Here is a table summarizing the different types of Green Accounting and their key features:

<u>Type of Green Accounting</u>	<u>Key Features</u>
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<u>Environmental Management Accounting (EMA)</u>	Focuses on internal decision-making and resource management by identifying, measuring, and analyzing environmental costs and benefits of a company's operations, and incorporating this data into management systems to improve environmental performance.
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<u>Full Cost Accounting (FCA)</u>	Calculates the total economic, social, and environmental costs of a product or service, including hidden costs that are not typically included in financial accounting, such as pollution and health impacts. This allows for a more comprehensive evaluation of the true cost of production and consumption and can inform policy and business decisions to reduce negative environmental impacts.
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<u>Sustainability Accounting</u>	Measures and reports on a company's economic, social, and environmental performance over time, using indicators such as energy and resource consumption, greenhouse gas emissions, and social impact assessments. Sustainability accounting aims to provide a holistic view of a company's sustainable development and can help identify areas for improvement and promote stakeholder engagement.
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Green Accounting Importance

Green accounting is important for several reasons:

Environmental Protection

Green accounting helps to identify the environmental impacts of economic activities and promotes sustainable development by encouraging the conservation and efficient use of natural resources.

Cost Savings

By measuring and managing environmental impacts, businesses can identify opportunities for cost savings through improved resource efficiency, reduced waste, and lower environmental compliance costs.

Risk Management

Green accounting helps businesses to identify and manage environmental risks that could impact their operations or reputation, such as regulatory changes or environmental disasters.

Stakeholder Engagement

By reporting on environmental and social performance, businesses can demonstrate their commitment to sustainability to stakeholders, including customers, investors, and regulators.

Policy Development

Green accounting can inform the development of policies and regulations that promote sustainable development and help to address environmental challenges such as climate change and biodiversity loss.

Green Accounting in India

Green accounting has gained importance in India in recent years, as the country faces numerous environmental challenges, including air pollution, water scarcity, and climate change. In response to these challenges, the Indian government has implemented various policies and initiatives to promote sustainable development and encourage businesses to adopt green accounting practices.

One such initiative is the National Green Accounting System (NGAS), which was launched by the Indian government in 2014. The NGAS aims to integrate environmental and economic accounting by incorporating environmental costs and benefits into the national accounts system. The NGAS also includes a set of environmental performance indicators to monitor progress toward sustainable development goals.

In addition, the Indian government has introduced various policies and regulations to promote environmental sustainability, such as the National Action Plan on Climate Change and the Swachh Bharat Abhiyan (Clean India Mission). These policies and initiatives have encouraged businesses to adopt green accounting practices, such as conducting environmental audits, reporting on environmental performance, and implementing environmental management systems.

Overall, green accounting is becoming increasingly important in India as businesses and policymakers recognize the need to balance economic growth with environmental sustainability....

n many markets, both in developed and developing countries, the call for increased transparency and accountability of the private sector is growing. Potential health and environmental risks posed by companies, and the goods and services they produce, are increasing pressure on them to generate, assess, and make information on their sustainability performance and impacts publicly available. In this sense Corporate Sustainability Reporting represents a potential mechanism to generate data and measure progress and the contribution of companies towards global sustainable development objectives as it can help companies and organizations measure their performance in all dimensions of sustainable development, set goals, and support the transition towards a low carbon, resource efficient, and inclusive green economy.

The Corporate Sustainability Reporting portfolio of work is embedded in UN Environment's Resources and Markets Branch and more particularly within the Consumption and Production Unit which main goal is the promotion of sustainable production and consumption policies, practices and initiatives through the engagement of governments, the businesses community and civil society organizations.

As part of the Corporate Sustainability Reporting agenda, the Consumption and Production Unit hosts the Secretariat of the Group of Friends of Paragraph 47 (GoF47), a government-led initiative formed in 2012 during the United Nations Conference on Sustainable Development (Rio +20) to advance recommendations put forward under paragraph 47 of the outcome document 'The Future We Want'. In addition, the Unit implements a number of activities including regional projects to facilitate the integration of sustainable practices by companies (SDG 12.6) in alignment with the Sustainable Development Goals and to facilitate governments' task of follow-up and review of the SDGs through higher quality and comparable information emerging from corporate reports.

A sustainability report is the disclosure of non-financial performance policies, methodologies and metrics to stakeholders, including investors, employees, customers and the public.

The practice encompasses environmental, social and governance (ESG) metrics. However, a sustainability report differs from an ESG report in that ESG evaluates the performance of companies against ESG metrics,

while sustainability looks broadly at the larger business model and methodologies.

More specifically, an ESG report uses a set of metrics to evaluate a company's ESG initiatives. Using various ESG reporting frameworks, companies can inform stakeholders of their positive impact on society and the environment, and of their corporate governance practices.

However, a sustainability report encourages more responsible and ethical business practices by considering the organization's relationship with the world around it. Sustainability is complex, so there are many reporting frameworks for companies, usually covering important topics like sustainable development, carbon emissions, supply chain and corporate social responsibility (CSR).

One of the main differences between ESG and sustainability is the notion of motivation versus outcomes. A sustainability report examines the business model or methodologies that motivate a company and its employees to act in the best interest of society. An ESG report is the measurement and outcome of those initiatives and provides both companies and investors with the ESG data needed to inform decision-making.

Similar to CSR, both ESG and sustainability reports help organizations improve trust and reputation among consumers by setting goals, meeting reduction targets and besting industry benchmarks. Companies can significantly improve decision-making and risk management by considering the three dimensions of sustainability in their larger sustainability strategy.

Guide7 sustainability trends to take action on

Take a look at the trends shaping the world of sustainable business and the insights that can help drive transformation.

What are the three dimensions of sustainability?

The three dimensions of sustainability are environmental, social and economic.

Environmental

Environmental sustainability focuses on addressing environmental issues such as climate change, greenhouse gas (GHG) emissions and

loss of biodiversity. One way for companies to reduce their environmental impact is to move away from finite resources like fossil fuels and natural gas and embrace renewable energy sources.

Social

Social sustainability is not yet as clearly defined. Some scholars say it embodies all human activity, and that all domains of sustainability fundamentally rely on social factors. This dimension prioritizes human rights and recognizes that the well-being of all people dictates the longevity, efficacy and sustainability of a society.

Economic

Economic sustainability means that businesses must be both profitable and environmentally conscious, which can sometimes conflict. However, companies are making progress in addressing climate issues. Chief sustainability officers, for instance, must reimagine value chains to improve ESG performance across entire global operations. That might mean a business must refine the way they source raw materials, adopt a fleet of electric vehicles or prioritize decarbonization initiatives. Regardless of the approach, the main goal for businesses remains to balance financial success with environmental protection.

Automated Accounting Processes,

Accounting automation refers to utilizing software to streamline and automate various accounting tasks, such as data entry, reconciliation, and financial reporting. It helps save time, reduce errors, and increase efficiency in managing financial processes.

Automation in accounting and finance has grown with AI and ML, enabling software to track, record, and customize data. By automating time-consuming repetitive tasks like data entry and calculations, it frees up time for the accounting teams to focus on other important responsibilities. While it doesn't eliminate the need for human involvement, it allows the accounting team to focus on more valuable and strategic activities, such as budgeting or investment planning.

Payroll

Automatically pays employees on time, deducts taxes and pension contributions, and generates employee statements

Expense management

Helps maintain accurate financial records, compliance, and agility

Accounts payable

Controls and manages financial transactions through a single system to reduce risk and mistakes

Invoice processing

Automatically converts and analyzes incoming invoices in a digital format to make it easier to track and correct errors

Reconciliation

Automatically compares a company's accounting records to its bank account data to save time and money

Cash flow management

Automatically schedules payments and tracks receivables to avoid late payments and penalties

Cloud-based Accounting

Cloud accounting is a method of storing and accessing financial data and accounting software online, instead of on a company's own servers or hard drives. It uses cloud computing and a Software as a Service (SaaS) model to allow users to access their accounts from anywhere with an internet connection.

Cloud accounting offers many benefits, including:

- **Accessibility:** Users can access their accounts and financial information 24/7, from any location.
- **Security:** Data is encrypted and stored on remote servers, and cloud service providers often offer security measures like multi-factor authentication.
- **Real-time information:** Users can access up-to-date financial information.
- **Automation:** Cloud accounting can automate tasks to reduce workloads.
- **Integration:** Cloud accounting software can integrate with a company's bank accounts, so transactions are automatically posted to the correct digital ledger.
- **Customization:** Some cloud accounting applications can be customized to meet the needs of a business

Cloud accounting software will continue to evolve and adapt to the changing needs of businesses, offering more advanced features and

integration capabilities. The future of cloud accounting holds great potential for increased efficiency, improved accuracy, and enhanced financial management for businesses of all sizes.

Data Analytics & Forecasting Tools

Data analytics are used by accountants to do things like discern patterns in customer spending, identify market behavior, anticipate trends and predict regulatory reactions.

Accountants who specialize in data analytics manage, analyze and mine multiple streams of data. Doing so provides them granular-level details that can be used to answer questions, identify patterns and make fact-based predictions.

Using data analytics, accountants can help a company:

- **Evaluate Performance:** The performance of every area of the business can be evaluated using predetermined metrics. An accountant may look at revenue data, quarterly goal performance or production numbers.
- **Mitigate Risk:** Areas of current or potential risk can be uncovered and managed in real-time. Funding needs, process flows and investment opportunities are all areas that an accountant may look at.
- **Understand Behaviors:** Tracking and reviewing consumer and internal behavior patterns and employee productivity waves enable accountants to drive business decisions and growth plans.
- **Build Business Plans:** With a detailed understanding of past and present business patterns, a company can confidently make plans for its future. An accountant would look at factors like historical sales numbers, employee retention patterns, organizational spending and equipment life-cycles.
- **Structure Business Improvements:** When an area of the company isn't performing up to expectations, data analytics can pinpoint where improvements might be needed. In order to create an effective strategy, an accountant may review sales forecasts, historical sales performance numbers and operating costs.
- **Find Opportunities:** Opportunities to grow and develop a competitive advantage can be uncovered by analyzing past performance and looking at current trends. These include operational capabilities, current customer breakdowns and market patterns.
- **Maximize Profits:** With the clear insight that data analytics can provide, a company can make decisions that will build up their bottom line. To help with this, accountants will look at a number of data points including past purchasing behaviors, current market trends, inventory management and customer orders.

Accounting firms can use data analytics and forecasting tools to analyze risk, make business decisions, and build forecasts. Some tools that can be used for data analytics and forecasting include:

- **Excel**

A popular tool for smaller businesses that's easy to use for accounting needs like building financial statements and budgets

- **Tableau**

A flexible tool that's good for visualizing data and working with large data sets

- **Power BI**

Combines business intelligence and data visualization, and can connect with other applications like Excel, Google Analytics, and Quickbooks

- **IDEA**

Software built for data analytics that allows users to import and analyze data quickly and efficiently

- **Python**

Offers flexibility and advanced capabilities

- **Monte Carlo simulation**

Useful for projects that require stochastic models to account for uncertainty

- **ARIMA**

Ideal for analyzing time series data to project future trends

- **Jirav**

A cloud-based platform that offers financial planning and analysis, and is customizable for tracking, forecasting, and sharing data

- **Oracle Hyperion**

An enterprise forecasting tool that combines planning, budgeting, and forecasting

Blockchain

Blockchain is a decentralized, distributed ledger that focuses on the ownership and transfer of assets. It records transactional data in a way that's almost impossible to manipulate.

Blocks of transactional data connect in chronological order. The chain of blocks gives the technology its name.

The blockchain database records the data of organizations and individuals across the world

Blockchain technology has the potential to replace the 500-year-old double-entry accounting system. Blockchain distributed ledger technology would popularize the triple-entry accounting system.

A Primer on General Ledgers and Double-Entry Accounting

Let's revisit the basics of a general ledger (GL) as we know it today. A GL includes all the assets, liabilities, equity, expense, and income ledgers, which make up a complete set of the financial transactions records. To make sure a GL is accurate, you'd use a double-entry accounting system.

In a double-entry accounting system, you record a debit and a credit of the same amount at the same time. In a triple-entry accounting system, a debit, credit, and a third entry is recorded. The third entry would be on the blockchain.

Standard accountancy requires a significant time investment from all organizations in the supply chain. Businesses keep their own ledger to ensure business' financial records are accurate and compliant. This process is also susceptible to human error.

Here's where blockchain's solution can make an impact.

How Will Blockchain Technology Affect the Accounting Industry?

There are three key aspects of blockchain that can affect the accounting industry.

- Smart contracts
- Decentralized, distributed ledger technology
- Easily verifiable financial records

Smart Contracts

A smart contract is one of many blockchain applications that can streamline tedious tasks in today's accounting.

With smart contracts, transactions automatically go through when certain conditions are met. This helps accounting professionals and organizations automate jobs like payroll and reconciliations. This would save organizations on costs linked to manual entry errors such as administrative expenses.

This application also helps clients and organizations against scams and fraud.

Decentralized, Distributed Ledger Technology

One of the first popular blockchain applications was that it cut out the middle man when transferring money. For example, you can send money peer-to-peer (P2P) without having to go through a credit card processor or bank.

Although the middle man slows down transactions and adds fees for their services, they're not all bad. The middle man plays a large role in protecting both parties in the exchange of assets from fraud. Blockchains maintain this security with public witnesses called miners. Miners replace a central authority's role in verifying transactions. This is done securely using a consensus protocol, or a set of rules based on mutual agreement.

Altering Transactions on the Blockchain

Blockchain's decentralized nature also helps act as proof that a transaction happened.

In the past, we'd use paper receipts for proof that a transaction occurred. Paper receipts were relatively hard to tamper with. With the introduction of digital payments came digital receipts, which are easier to tamper.

Blockchain tech is again a promising solution to this problem.

Blockchain's immutable nature comes from the fact that once a public consensus validates a transaction into the blockchain, it's virtually impossible to alter or delete the transaction.

If an organization modifies a transaction's data in the blockchain, it'll affect the hash value. This will be an immediate red flag that someone tampered with the data. Passwords commonly use hashes too.

A hash value is basically a generated string of characters. It protects the sensitive data of the transaction and acts as a receipt that verifies the transaction occurred at a certain time.

AI in Accounting

AI in accounting refers to the use of artificial intelligence (AI) technologies, such as machine learning algorithms and natural language processing, to automate and enhance various accounting processes.

This includes tasks such as financial reporting, audit and compliance, fraud detection, and data analysis.

AI in accounting can help improve accuracy and efficiency, reduce costs, and provide valuable insights and predictions for decision-making.

Let us take a look at the benefits of Artificial Intelligence (AI) in accounting and our daily lives.

1) Increased Efficiency and Productivity

AI makes it easier to do repetitive accounting tasks, which frees up time for more important tasks. This makes workers more productive and efficient because they can focus on what they do best.

2) Automated Tasks and Reduced Human Error

The use of automated tasks in accounting and finance has the potential to improve the accuracy and efficiency of financial processes significantly.

The potential for human error is decreased when routine and repetitive processes are automated, which ultimately results in more reliable financial data and reporting.

3) Improved Customer Experience

Using ChatGPT to draft emails can save you time. Currently, my team and I are utilizing AI for customer support.

4) Enhanced Data Analysis and Predictions

AI algorithms are capable of quick data analysis. This makes AI an effective tool for predicting and identifying trends.

AI helps individuals and businesses manage financial data efficiently by analyzing transactions, cash flow, budget, and accounting data to identify trends, errors, and areas of improvement.

Cost Savings

AI's process automation can save accounting firms time and money by automating routine tasks, freeing up workers to focus on strategic initiatives, and enabling businesses to increase productivity and profits while cutting expenses.

Big Data in Accounting

Big Data accounting and finance is a collection of vast sets of unstructured data stored in various forms gathered from different sources. It comes in at such a fast rate that a traditional server cannot process it and is usually measured in terabytes and zettabytes. Big data in accounting and data science enables accountants to identify issues with real-time access to the data so that businesses can make decisions based on hard evidence and facts instead of uncertain assumptions. This is what has made understanding big data analytics in accounting meaning for accounting and finance professionals a must today. The Journal of big data and accounting data science journals are key tools in understanding the realm of data science and data analytics. Big data comes in complex data sets, which can be broken down into three defining properties. Collectively, they are called the three Vs of Big data in Accounting.

1. Volume

The first and most prominent property is volume. The data is voluminous to an extent to reach unparalleled heights. It was estimated that approximately 2.5 quintillion bytes of data are created each day. As a result, more than 40 zettabytes of data could be produced by 2021. This brings to light an increase of 300 times from 2005. And so, it's now common for big data firms to have Terabytes and even Petabytes of data stored on drives and servers.

2. Velocity

The fast growth of data has changed the way we see it. There once was a time when we didn't know the importance of data in the corporate world and accounting science. However, with the evolution of how it is gathered, many huge businesses now rely on it daily. Velocity measures how fast the data is coming in. Some data comes in real-time; some comes in fits and starts, sent in batches. And as not all platforms experience the incoming data at the same pace or speed, it's important to not jump to conclusions without having all the facts and figures.

3. Variety

Earlier, data used to be collected from one place and delivered in a single format. With time, it has evolved from database files - such as Excel, and CSV, and access to now being presented in nontraditional forms, like text, video, pdf, graphics on social media, and through tech, like wearable devices. Though this data is beneficial, it does add to the work and requires advanced analytical skills to decode, make it manageable and work. Now that you have understood what exactly big data is, let's see how big data has influenced accounting and data science.

Impact of Big Data on Accounting and Data Science

The rapid growth of technology and high-volume data generation is changing how industries and businesses operate. Traditionally, humans examined and analyzed numbers, and decisions were made based on conclusions drawn from calculated risks

and trends. However, computers have usurped this functionality, and it is, now, done through data mining accounting. As a result, the market for big data technology in finance offers excessive potential. Here is how big data has transformed accounting and data science:

1. Real-Time Access to Data

Previously, manual data recording methods limited the visibility of data. It used to be challenging to complete the tasks every month before the books were closed. Today, you have real-time access to the accounting data, which helps you instantly correct errors in reports, escalate efficiency, and save both time and money. Understanding big data analytics accounting and utilizing it to analyze reports can help you make better business decisions and set performance benchmarks. If you don't know what is big data analytics in accounting, then now you understand it.

2. Applying Data Analytics to Big Data

Big data comes in a large volume of unstructured data that must be organized. Applying data analytics to big data in accounting helps businesses gain significant insights, predict future possibilities, and automate financial tasks. For an accountant to be a big data expert, they must learn technical and analytical skills to manage statistics and examine large data sets with data mining accounting.

3. Data Analytics and Audit

The change in data auditing is one of the best examples of big data in accounting. Today, auditing is so much more than big data analysis in accounting of the income statement and balance sheet. Auditors might have had to manually go through big files to track risks in the past. However, the rise of technology and big data in accounting has revolutionized how auditing is done. You can now easily assess large volumes of unstructured data in the form of emails, company records, and more by using computers, algorithms, and other technology.

4. Better Risk Management

Today, it is crucial for big firms to regularly identify risks and their effects on the business. Risks in financial services are mostly related to mergers and acquisitions, frauds, supply-chain risks, etc. With the introduction of big data in accounting, you can significantly improve risk management through advanced customer behaviour analysis, foresee shifts in economic trends, and much more. Moreover, it can also help analyze card transaction patterns. This includes the amount, timing, location, and more that can help identify swindling transactions and block the card. In the case of liquid risk management, big data groups can give in-depth, valuable insights into the cash flow, which can be used to improve liquidity management. The sooner the accountants learn to identify these risks, the better is the chance of alleviating them.

5. Improved Insights

Big data analytics in accounting allows for the collection and analysis of vast amounts of financial data from various sources. This enables accountants to gain deeper insights into a company's financial performance. They can identify trends, patterns, and anomalies that might go unnoticed with traditional methods. These insights can inform strategic decisions, helping businesses allocate resources more effectively and optimize their financial strategies.

6. Predictive Modeling

Big data enables the development of predictive models that can forecast future financial outcomes with greater accuracy. By analyzing historical data and external factors, accountants can create models that predict cash flow, revenue, and expenses. This helps in provide measures.

7. Fraud Detection and Prevention

Big data analytics is a powerful tool in detecting and preventing financial fraud. By monitoring large volumes of financial transactions in real time, anomalies or suspicious patterns can be quickly identified. This proactive approach helps in fraud prevention rather than merely reacting after an incident. It safeguards a company's assets and reputation, reducing the financial impact of fraudulent activities.

Cost Efficiency: Big data technologies can automate many routine accounting tasks, such as data entry and reconciliation. This not only reduces the risk of human error but also frees up accountants to focus on more strategic activities. It leads to cost savings for businesses while improving the accuracy of financial reporting.

8. Compliance and Regulatory Reporting

Big data analytics assists in ensuring compliance with complex financial regulations. By continuously monitoring and analyzing financial data, accountants can identify discrepancies and ensure that the company's financial reporting aligns with regulatory standards. This minimizes the risk of penalties and legal issues. active financial planning and risk management. For instance, companies can better anticipate periods of financial stress and take preventive measures.

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